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Low-energy control of electrical turbulence in the heart

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Controlling the complex spatio-temporal dynamics underlying life-threatening cardiac arrhythmias such as fibrillation is extremely difficult, because of the nonlinear interaction of excitation waves in a heterogeneous anatomical substrate[1–4]. In the absence of a better strategy, strong, globally resetting electrical shocks remain the only reliable treatment for cardiac fibrillation[5–7]. Here we establish the relationship between the response of the tissue to an electric field and the spatial distribution of heterogeneities in the scale-free coronary vascular structure. We show that in response to a pulsed electric field, E, these heterogeneities serve as nucleation sites for the generation of intramural electrical waves with a source density $\rho(E)$ and a characteristic time, τ, for tissue depolarization that obeys the power law $\tau \propto E^{\alpha}$. These intramural wave sources permit targeting of electrical turbulence near the cores of the vortices of electrical activity that drive complex fibrillatory dynamics. We show *in vitro* that simultaneous and direct access to multiple vortex cores results in rapid synchronization of cardiac tissue and therefore, efficient termination of fibrillation. Using this control strategy, we demonstrate low-energy termination of fibrillation *in vivo*. Our results give new insights into the mechanisms and dynamics underlying the control of spatio-temporal chaos in heterogeneous excitable media and provide new research perspectives towards alternative, life-saving low-energy defibrillation techniques.

Spatially extended non-equilibrium systems display spatio-temporal dynamics that can range from ordered to turbulent. Controlling such systems is one of the central problems in nonlinear science and has far-reaching technological consequences. Few examples of successful control with applications in physics and chemistry have been demonstrated[8,9]. In biological excitable media, the systems' complexity makes successful control challenging. This difficulty applies in particular to electrical turbulence in cardiac tissue, known as fibrillation. During fibrillation, synchronous contraction of the muscle is disrupted by fast, vortex-like, rotating waves of electrical activity[1–4]. At the core of the vortex is a line of phase singularities called a filament. It is known that vortex instabilities and interactions[10,11] lead to self-organized, turbulent electrical dynamics. In the heart, electric turbulence arises in electromechanically anisotropic and heterogeneous cardiac muscle, which has complex geometry[12]. As we show below, this natural complexity can be used as a substrate for successful control of electrical turbulence.

The physiological mechanisms underlying the dynamics and control of electrical turbulence remain largely unknown[13]. The only clinically effective method for eliminating vortices in the heart is the delivery of a high-energy electric shock that both depolarizes and hyperpolarizes the tissue with a voltage gradient of about $5\,\mathrm{V\,cm^{-1}}$. When applied externally, this shock can be as large as 360 J (1 kV, 30 A, 12 ms)[5]. Although defibrillators that use this approach are used routinely in emergency medicine, treatments are often associated with severe side effects[6,7,14].

Here we provide a new understanding of the biophysical mechanisms involved in the control of cardiac fibrillation and demonstrate low-energy control and termination of cardiac fibrillation *in vivo*. Two internal catheters with coiled wire electrodes were inserted into the right and left atria of adult beagle dogs (Fig. 1a) and sustained atrial fibrillation (AF) was induced (see Methods). We compared the energies required for defibrillation using a single, high-energy shock (standard defibrillation) and a sequence of five low-energy electric field pulses (low-energy antifibrillation pacing (LEAP)). A representative time series for successful LEAP termination via a monophasic action-potential electrode inserted into the right atrium is shown in Fig. 1b. For $t < 0$, the signal shows irregular oscillations with a dominant frequency f_v of 6.8 ± 0.1 Hz. The control interval starts at $t = 0$ (Fig. 1b, grey shaded area) and after control, the arrhythmia is terminated and normal sinus rhythm is restored. In this example, the energy required for terminating AF was 0.074 ± 0.012 J, seven times less than the energy needed for standard defibrillation in this preparation (0.52 ± 0.20 J). This substantial reduction was reproduced in 56 episodes in seven *in vivo* experiments, in which we found an average energy reduction of 84% (see Fig. 1d) ($P < 10^{-7}$).

To identify the biophysical mechanisms underlying LEAP, we conducted *in vitro* experiments with isolated, perfused atria, using the same electrode configuration as in the corresponding *in vivo* experiments. *In vitro*, fluorescence imaging allowed quantitative measurements of the propagation of action potentials on the surface of the tissue, with high spatial and temporal resolution (see Methods). To assess differences in LEAP effectiveness between the *in vivo* and *in vitro* preparations, three of the five *in vitro* preparations were derived from the hearts used in the *in vivo* experiments. A time series of the fluorescence signal is shown in Fig. 1c (same heart as in Fig. 1b). For $t < 0$, we observed sustained AF with a dominant frequency of $f_v = 6.8 \pm 0.1$ Hz. After LEAP (Fig. 1c, grey shaded area), normal sinus rhythm was restored. The LEAP pulse energy was 0.066 ± 0.017 J, compared to 1.15 ± 0.29 J for standard, single-pulse defibrillation. The overall energy reduction in the *in vitro* experiments ($n = 5$ preparations, 39 defibrillation episodes and 46 LEAP episodes) was 91% (Fig. 1d) ($P < 10^{-7}$). There was no significant difference in the energy required *in vivo* and *in vitro* for LEAP ($P = 0.49$) and conventional defibrillation ($P = 0.63$). Our findings are also in agreement with a separate set of *in vitro* experiments[15] in canine right-atrial preparations ($n = 8$), in which LEAP terminated AF with a success rate of 93%, using only 13% of the energy per pulse required for a single shock

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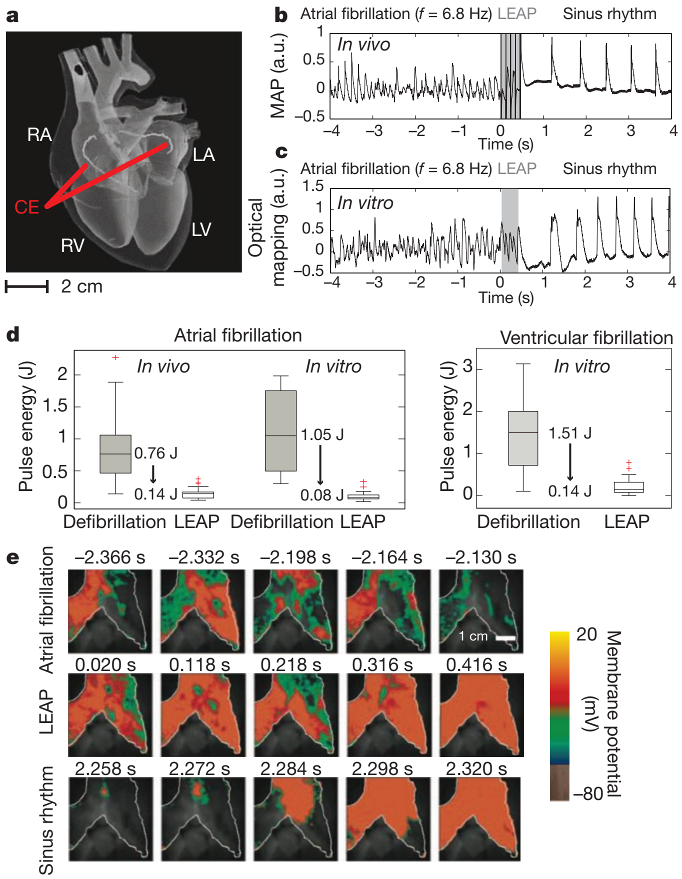

Figure 1 | Low-energy termination of cardiac electrical turbulence *in vivo* and *in vitro*. **a**, Schematic of the anatomy of the heart. LA, left atrium; LV, left ventricle; RA, right atrium; RV, right ventricle. A pulsed electric field was applied with standard cardioversion coiled wire electrodes (CE) inserted into the left and right atria by catheters (see Supplementary Information). **b**, Monophasic action potential (MAP) recording of termination of AF using LEAP *in vivo*. a.u., arbitrary units. Dominant frequency $f_v = 6.8 \pm 0.1$ Hz, $n = 5$ pulses, pulse duration $\Delta t = 8$ ms, pacing cycle length $T_p = 99$ ms, pulse energy $W = 0.074 \pm 0.012$ J. **c**, Termination of AF *in vitro*, measured from the atrial epicardium of the same heart as in **b**, by optical mapping (see **e**). The signal from a 0.3×0.3 mm^2 region is shown ($f_v = 6.8 \pm 0.1$ Hz, $n = 5$, $\Delta t = 8$ ms, $T_p = 90$ ms, $W = 0.066 \pm 0.017$ J). **d**, Reduction in pulse energy using LEAP versus standard defibrillation. *In vivo* AF ($n = 7$): LEAP (56 episodes, mean energy $\bar{W} = 0.14 \pm 0.08$ J); defibrillation (22 episodes, $\bar{W} = 0.89 \pm 0.56$ J). *In vitro* AF ($n = 5$): LEAP (46 episodes, $\bar{W} = 0.10 \pm 0.07$ J); defibrillation (39 episodes, $\bar{W} = 1.15 \pm 0.58$ J). *In vitro* ventricular fibrillation ($n = 7$): LEAP (28 episodes, $\bar{W} = 0.17 \pm 0.16$ J); defibrillation (12 episodes, $\bar{W} = 1.34 \pm 0.89$ J; see Supplementary Information). Box plots show the median and the 25th and 75th percentiles. Whiskers indicate the statistically significant data range and red crosses mark outliers. **e**, Optical mapping of the AF termination that is also shown in **c**. During AF, complex spatio-temporal propagation of electrical excitation waves was observed (white line indicates boundary of atrium). LEAP ($n = 5$, $\Delta t = 90$ ms) progressively synchronized the tissue (see Supplementary Movies 1 and 2). Data are given as mean ± s.d. unless stated otherwise.

($P < 0.002$). Furthermore, LEAP effectiveness was demonstrated for ventricular fibrillation *in vitro* ($n = 7$ canine preparations, 12 defibrillation episodes and 28 LEAP episodes). In these experiments (Fig. 1d), the average energy reduction of LEAP versus a single shock was 85% ($P < 10^{-5}$). The spatio-temporal excitation dynamics of the right atrium *in vitro* before, during and after control are shown in Fig. 1d (see also Supplementary Movie 1). During fibrillation, waves of turbulent electric activation propagate across the atria. At $t = 0$, a sequence of five electric pulses is applied at the coil electrodes, followed by a transient, spatio-temporal reorganization of the activation waves. After each pulse, the area that is activated increases, indicating progressive synchronization of the myocardium; fibrillation then terminates and normal sinus rhythm (Supplementary Movie 2) can resume.

To elucidate the mechanism of defibrillation by LEAP, we studied the response of quiescent atrial and ventricular tissue to a homogeneous, pulsed electric field (Fig. 2a, b). In Fig. 2c, images taken at 1.5 ms, 3 ms and 6 ms after the pulse (0.22 V cm^{-1}) show depolarization induced by a single source. However, with increasing electric field strengths of 0.22 V cm^{-1}, 0.39 V cm^{-1} and 0.50 V cm^{-1}, the number of sources increases to several dozen over the entire tissue. The locations of these sources and the wave propagation patterns are summarized in the isochronal maps shown in Fig. 2d. The density of sources, shown in Fig. 2c and d, increased with increasing field strength for both the ventricle and the atrium, thereby decreasing the activation time (Fig. 2b).

The results can be explained in the context of virtual electrodes[15–20]. In the bidomain representation, the voltage in cardiac tissue is the potential drop between the intracellular and extracellular medium. Theory predicts[16] that, in the presence of an electric field, discontinuities in tissue conductivity, such as blood vessels, changes in fibre direction, fatty tissue and intercellular clefts, induce a redistribution of intracellular and extracellular currents that can locally hyperpolarize or depolarize the cells. At the depolarization threshold, an excitation wave is emitted[15–17,21].

The electric field that is necessary to produce an activation, as a function of the size of the conduction discontinuity in quiescent tissue,

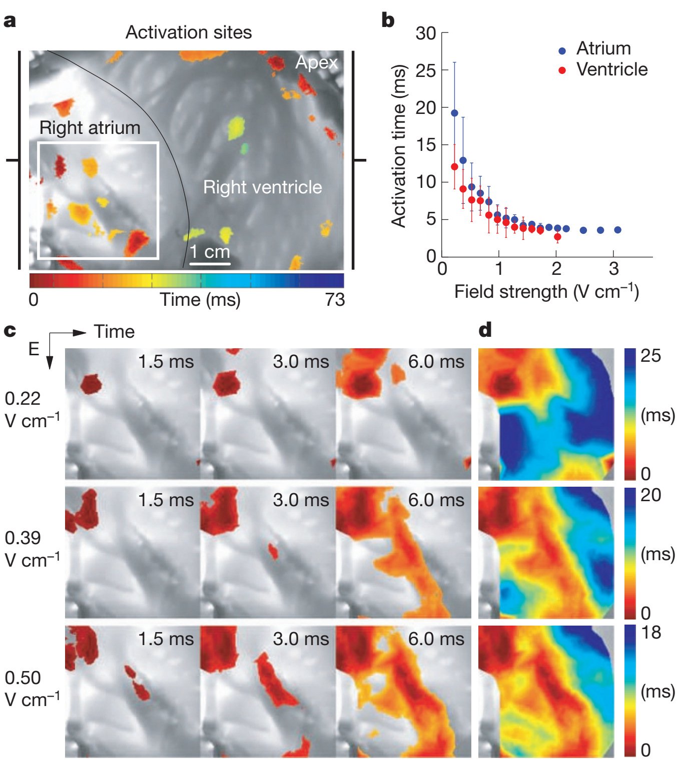

Figure 2 | Sites of activation in a cardiac preparation. **a**, Canine wedge preparation (7.5×5.6 cm^2) consisting of right atrium and right ventricle. At $t = 0$ s, an electric field pulse of strength $E = 0.34$ V cm^{-1} was applied for 5 ms. The colour indicates the time of local activation observed with fluorescence imaging on the endocardium; the greyscale trans-illumination image shows the complex anatomy of the endocardium. The white square marks the area shown in panel **c**. **b**, Mean activation times $\tau(E)$ for atria (blue circles, $n = 3$ preparations, 17 measurements of $\tau(E)$) and ventricles (red circles, $n = 6$ preparations, 24 measurements of $\tau(E)$) in response to an electric field pulse of strength E and duration 5 ms. Error bars indicate s.d. **c**, Activation of the atrium (in the region indicated by the white square in panel **a**) after an electric field pulse at $t = 0$. With increasing field strength, the number of activation sites increased and the time interval for total activation decreased. The colour code for each row is shown in **d** (see Supplementary Movie 3). For $E < 0.2$ V cm^{-1}, no waves were observed. **d**, Isochronal maps of the activation sequences shown in **c**.

can be estimated by approximating the discontinuous geometries (circles in two dimensions, spheres in three dimensions) and linearizing the bidomain model equations[16,21,22] around the resting membrane potential[21], obtaining

$$\nabla^2 e - \frac{e}{\lambda^2} = 0 \tag{1}$$

where $e = \Phi - \Phi_{rest}$, Φ and Φ_{rest} are the induced and resting membrane potentials and λ (~0.35 mm) is the space constant. For a non-conducting tissue region of radius R, the minimum electric field, E, necessary to bring the voltage above the excitation threshold Φ_t at the boundary ($r = R$) is given by

$$E = -\frac{\Phi_t - \Phi_{rest}}{\lambda} \frac{K_1'(R/\lambda)}{K_1(R/\lambda)} \tag{2}$$

where $K_1(R/\lambda)$ is the modified Bessel function of the second kind (see Supplementary Information). Qualitatively, equation (2) indicates that the smaller the size of the heterogeneity, the larger the electric field (E) required to emit waves. Equivalently, when E increases, discontinuities with a size larger than $r \geq R_{min}(E)$ are recruited as wave sources, where $R_{min}(E)$ is obtained by solving equation (2). $R_{min}(E) \approx 1/E$ when E is large (Supplementary Fig. 13). The heart has heterogeneities of all sizes R, with a distribution $p(R)$. Assuming that these heterogeneities are uniformly distributed in the tissue, the density of recruited wave sources is

$$\rho(E) = \frac{N_{total}}{V} \int_{R_{min}(E)}^{R_{max}} p(R)\mathrm{d}R \tag{3}$$

where R_{max} is the size of the largest discontinuity in the tissue.

To quantify $p(R)$ associated with blood vessels in intact cardiac tissue, the coronary arteries of eight cardiac preparations were perfused with contrast agent and scanned using micro-computed tomography (micro-CT; Supplementary Figs 1–3 and 12). As shown in Fig. 3a and e, the size distribution $p(R)$ of discontinuities yielded power laws $p(R) \propto R^{\alpha}$ with exponents $\alpha = -2.74 \pm 0.05$ ($n = 5$ preparations, Supplementary Figs 4 and 6 and Supplementary Table 1) for atria and $\alpha = -2.75 \pm 0.30$ ($n = 3$ preparations, Supplementary Figs 4 and 5 and Supplementary Table 2) for ventricles. In biological systems, power-law scaling reflects generic underlying physical and physiological design principles relating form and function[23,24].

The geometric structure of the coronary vasculature results in characteristic activation dynamics in response to a pulsed electric field, so the activation times as a function of field strength that are shown in Fig. 2b can be predicted using equation (3) and $p(R)$. The excitation wave emitted by a single heterogeneity and propagating radially at constant velocity v excites a volume $V(\tau) = 4\pi(v\tau)^3/3$ in a time interval τ. For N heterogeneities, uniformly distributed with density $\rho = N/V$, the entire tissue is excited after

$$\tau(\rho) = (3/(4\pi\rho))^{1/3}/v \tag{4}$$

Equations (2) to (4) quantitatively relate structural properties (that is, the size distribution) to functional dynamics (that is, the activation times). In particular, these equations indicate (see Supplementary Information) that the exponents of size distributions, $p(R) \propto R^{\alpha}$, and activation times, $\tau \propto E^{\beta}$, are related, for large E, by

$$\beta = (1+\alpha)/d \tag{5}$$

where d is the geometric dimension of the tissue. To assess the role of tissue geometry, we measured the scaling exponents α and β for the thick ventricular wall ($d = 3$) and the thin, quasi-two-dimensional atrial wall ($d = 2$). We used the measured size distribution exponents, α, to estimate the activation time exponents, β_p, using equation (5). The predicted exponents β_p were found to be in good quantitative agreement with the observed exponents β_o for atrial and ventricular tissue (Table 1, Supplementary Figs 7–9 and Supplementary Tables 3 and 4).

From the measured size distributions $p(R)$ shown in Fig. 3a, we numerically estimated the intramural wave source densities $\rho(E)$ using

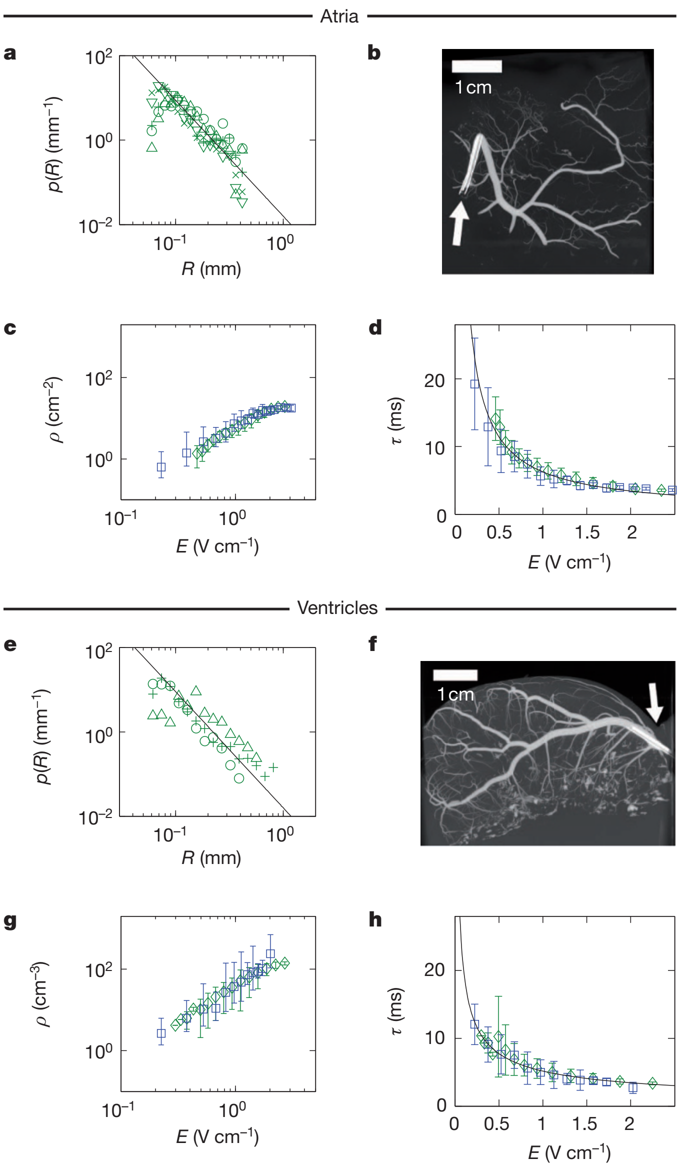

Figure 3 | From anatomical structure to activation dynamics in atria (a–d) and ventricles (e–h). **a**, Probability distribution of radii, $p(R)$, of canine coronary arteries in atrial tissue, obtained from micro-CT measurements (symbols represent $n = 5$ different preparations, for details see Supplementary Information). The black line indicates the power law $p \propto R^{-2.74 \pm 0.05}$ (mean scaling exponent of all preparations). **b**, **f**, Examples of atrial (**b**) and ventricular (**f**) anatomical structure of coronary arteries (see Supplementary Movies 9 and 10). White arrows indicate the position of the catheters used to inject the contrast agent (see Methods). **c**, Density of wave sources derived from equation (3) and from the atrial measurements shown in panel **a** (green diamonds), and corresponding density estimated from the activation-time measurements shown in panel **d** (blue squares). The predicted density from the structural data in **a** is plotted as the mean of the predictions from individual preparations. **d**, Atrial activation-time measurements using optical mapping (blue squares), and corresponding prediction of activation dynamics (green diamonds), on the basis of the source density obtained in **c** from the size distribution in **a** (plotted as the mean of predictions from individual preparations). The black line indicates the power law $\tau \propto E^{-0.87 \pm 0.03}$ (see Table 1 and Supplementary Information). **e**, Probability distribution $p(R)$ of coronary artery radii for ventricular tissue ($n = 3$). The black line indicates the power law $p \propto R^{-2.75 \pm 0.3}$ (mean scaling exponents of all preparations). **g**, Density of wave sources derived from the ventricular measurements shown in **e** (green diamonds), and corresponding density estimated from activation-time measurements shown in **h** (blue squares). **h**, Ventricular activation-time measurements (blue squares) and prediction of activation times (green diamonds) based on $p(R)$. The black line indicates the power law $\tau \propto E^{-0.58 \pm 0.10}$ (see Table 1 and Supplementary Information). Error bars indicate s.d. in all plots. Additional three-dimensional illustrations and animations are available at http://thevirtualheart.org/vessels.

equation (3) (green symbols in Fig. 3c), and $\tau(E)$ using equation (4) (green symbols in Fig. 3d, see Supplementary Information). Similarly, from the measured activation time $\tau(E)$ shown in Fig. 3d (blue symbols), we numerically estimated the intramural wave source density $\rho(E)$ (blue symbols in Fig. 3c) using the inverse of equation (4). Figure 3c, d, g and h shows comparisons between measured and calculated data for atria and ventricles, respectively. Green symbols represent results based on the measured size distribution $p(R)$, whereas results based on the measured activation time $\tau(E)$ are shown in blue. Again, excellent agreement was found (Table 1). Our results show that the structural properties of the coronary vasculature quantitatively describe the timescales of tissue activation.

The experimental results shown in Fig. 3 indicate that, in addition to the known effect of recruiting tissue boundaries during an electric shock[25], the heterogeneity of the vascular structure also allows the recruitment of an effective density of wave sources inside the myocardium. These distributed wave sources can be used for non-invasive, intramural multi-site pacing, which is key to the novel approach of LEAP, as demonstrated in Fig. 1. Previous studies aimed to control a single, isolated vortex attached to a large heterogeneity[21,26,27] and it was thought that in tissue with many heterogeneities of different sizes, it should be possible to control disordered regimes[28] by varying the electric field strength to control the number of wave sources[29]. An experimental demonstration of this approach was provided for AF[15]; however, the underlying control mechanism was not explored.

As shown in Fig. 3, strengthening the electric field increases the density of wave sources. Thus, the probability of wave sources, both in regions of excitable tissue and in the vicinity of filaments, increases. Being excitable, these regions can be brought above the excitation threshold and fully depolarized by the applied electric field pulse. Thus, they act as control sites that directly affect vortex filaments, which are the source of fibrillation.

Figure 4 provides experimental evidence that LEAP interacts directly with multiple vortices simultaneously. During AF, complex spatio-temporal dynamics with several interacting waves are observed (Fig. 4e and Supplementary Movie 10), evidenced by the presence of multiple phase singularities. During the episode of AF shown in Fig. 4, we observed 9.9 ± 4.4 (mean $\pm$ s.d.) phase singularities, resulting in the spatial complexity of the dominant frequency map (Fig. 4d, inset) and the corresponding broad probability distribution of frequencies, centred around 15 Hz (Fig. 4d). The observed spatio-temporal complexity can also be found in the pseudo-electrocardiogram (pseudo-ECG) and the percentage of area activated (PAA) (Fig. 4a, b and Supplementary Information). Whereas both signals show complex amplitude fluctuations during AF, the periodic perturbations during LEAP defibrillation result in increasingly coherent dynamics in the entire tissue, associated with a progressive increase in amplitude of the pseudo-ECG and the PAA. During AF, the normalized PAA is 0.28 ± 0.11 (mean $\pm$ s.d.), whereas during LEAP, PAA increases with each pulse towards one, that is, simultaneous activation of the entire tissue. After LEAP, AF is terminated and periodic, normal rhythm resumes (Supplementary Movie 10).

Figure 1 demonstrates the efficacy of LEAP defibrillation and Fig. 4 shows that tissue synchronization, and therefore control of fibrillation, is achieved by perturbing the system near the filaments, where it is most susceptible to perturbations[30]. A periodic perturbation can reach a nearby filament only if $f_{LEAP} > f_v$, where f_{LEAP} is the frequency of the perturbation and f_v is the frequency of the associated vortex. However, if f_v exceeds f_{LEAP}, a distant wave source cannot perturb the filament, owing to wave annihilation (Supplementary Figs 10 and 11). Nevertheless, as shown in Fig. 4a and b, AF was terminated with $f_v > f_{LEAP}$. We quantified termination of multiple filaments with $f_v > f_{LEAP}$ in atrial tissue ($n = 4$ preparations, 6 episodes) and ventricular tissue ($n = 1$ preparation, 3 episodes) (Supplementary Table 5). This result demonstrates simultaneous, local access to multiple filaments. Our experiments indicate that by adjusting the number of recruited sites, this approach may successfully terminate fibrillation regardless of whether fibrillation is caused by multiple re-entrant waves or a single mother rotor[1–3]. Consequently, this mechanism is applicable to multiple underlying causes of fibrillation, in both atria and ventricles. Our findings on the mechanism of defibrillation,

Table 1 | Observed and predicted scaling exponents

Tissue	d	α_o	β_o	β_p†	β_p*
Atrium	2	-2.74 ± 0.05	-0.81 ± 0.23	-0.88 ± 0.20	-0.87 ± 0.03
Ventricle	3	-2.75 ± 0.30	-0.75 ± 0.18	-0.74 ± 0.25	-0.58 ± 0.10

$p(R)$ and the activation time $\tau(E)$ for atria and ventricles. Statistical analysis showed no significant difference for α_o for atria and ventricles (see Supplementary Information).
* β_p as obtained from the high-field-strength approximation equation (5).
† Average of β_p obtained from least squares fits to activation times, obtained from direct numerical estimation using $p(R)$ and equations (2) to (4) (Supplementary Tables 1–4).

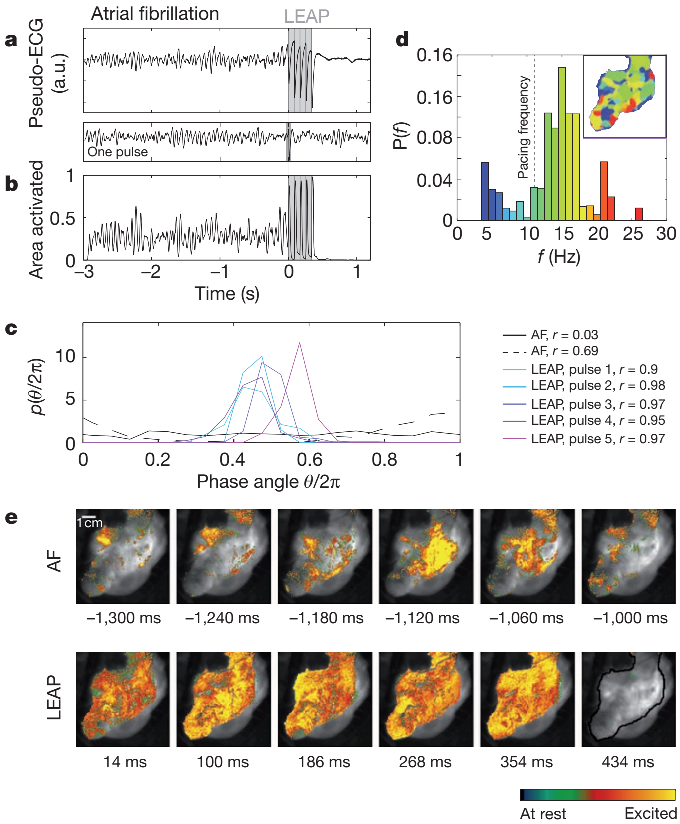

Figure 4 | Direct access to vortex cores. **a**, Termination of fibrillation with LEAP. The pseudo-ECG is obtained from the optical mapping experiments (see Supplementary Information). The dominant frequency during AF is $f_v = 15.0 \pm 0.5$ Hz (see inset in **d**). The vertical grey lines indicate the times at which five LEAP pulses ($f_{LEAP} = 11.8$ Hz, $E = 1.96$ V cm^{-1}) were delivered. A single pulse at this field strength did not terminate fibrillation (see pseudo-ECG below panel **a**). After LEAP, normal rhythm was resumed. The termination of fibrillation with $f_{LEAP} < f_v$ was observed in atria ($n = 4$ atria, 6 episodes) and ventricles ($n = 1$ ventricle, 3 episodes). **b**, The area activated indicates tissue synchronization during LEAP. Grey lines indicate timing of LEAP pulses. **c**, Probability distribution of phases θ_j during AF, and for the times of strongest synchronization (maximum order parameter r, see legend) after each LEAP pulse (see Supplementary Information). During AF, broad phase distributions indicate partial coherence (solid and dashed black lines show two representative distributions). LEAP with $f_{LEAP} < f_v$ induces synchronization and thus termination of AF. **d**, Probability distribution of dominant frequencies during AF, obtained from optical mapping. The dominant frequency map (inset) shows a complex spatial domain structure corresponding to multiple interacting waves (for colour code see histogram; $f_v = 15.0 \pm 1.0$ Hz; $f_{LEAP} = 11.8 \pm 0.5$ Hz, indicated by a vertical dashed line). **e**, Spatio-temporal dynamics during AF, LEAP (images taken at the times of maximum-area-activated after each pulse; see **b**) and sinus rhythm (see Supplementary Movie 8). The greyscale image shows the atrium (see black line in the last image; 70×70 mm^2). The last image shows quiescence after AF termination.

together with the *in vivo* proof of defibrillation of atria with LEAP, should allow the development of new approaches towards alternative life-saving defibrillation techniques.

METHODS SUMMARY

Experiments were conducted in open-chest anaesthetized dogs or in isolated, arterially perfused canine atrial and ventricular preparations *in vitro* (see Methods). Atrial fibrillation was induced *in vivo* using a pace-down protocol during stimulation of the right vagus nerve (2–4 mA at 10 Hz), and was induced *in vitro* during perfusion with acetylcholine (1–3 μM). Ventricular fibrillation was induced *in vitro* using rapid pacing. Membrane potential was recorded *in vitro* by optical mapping using a voltage-sensitive AminoNaphthylEthenylPyridinium dye (di-4-ANEPPS). Standard 6.5F cardioversion catheters were used to deliver defibrillation shocks *in vivo* and *in vitro*. The shocks consisted of 1–5 symmetrical biphasic pulses of 8-ms duration at shock strengths of 20–100 V, delivered via a custom-built cardioverter/defibrillator. Immediately after the optical mapping experiments, tissues were injected with 1–2 ml of Microfil contrast agent at 0.05–0.15 ml min^{-1} via the same cannula used for perfusion. The chambers were then filled with silicone to preserve tissue morphology during scans performed using a GE 120 micro-CT scanner with 25-μm *x*–*y*–*z* resolution, to determine blood vessel sizes and distributions.

Full Methods and any associated references are available in the online version of the paper at www.nature.com/nature.

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Supplementary Information is linked to the online version of the paper at www.nature.com/nature.

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METHODS

Cardiac preparation *in vivo*. Adult beagle dogs ($n = 7$) were induced with fentanyl citrate ($0.05\,mg\,kg^{-1}$) and propofol ($0.025\,mg\,kg^{-1}$), given as two sequential intravenous boluses. After intravenous administration of atracurium ($0.4\,mg\,kg^{-1}$), the dogs were intubated, artificially ventilated with oxygen and maintained under general anaesthesia with a continuous-rate infusion of fentanyl ($0.004\,mg\,kg^{-1}\,h^{-1}$). A median sternotomy was performed and the heart was suspended in a pericardial cradle. Normal body temperature was maintained with an inflatable heated body jacket and fluid losses were replaced with lactated Ringer's solution (at $10\,ml\,kg^{-1}\,h^{-1}$, intravenously). Blood oxygen saturation and carbon dioxide levels were monitored continuously and maintained within normal limits by varying the tidal volume and/or respiratory rate.

8 F introducers were placed in both femoral veins to allow the passage of catheters that would provide programmed stimulation to induce AF and to suppress AF using LEAP, as well as to monitor atrial activation. The catheters were 6.5 F and 160–175 cm long, with a single coiled-wire monofilament cardioversion electrode 6 cm in length situated 1 cm from the tip of the catheter (Modified Rhythm and Impact catheters, Rhythm Technologies Inc.). The catheters also carried two sensing electrodes distal to the coiled wire electrode. One catheter was placed via a femoral vein introducer and advanced until its coil electrode was positioned in the left pulmonary artery. A second catheter was placed so that its coil electrode was positioned in the right atrium. Alternatively, one of the catheters was inserted into the left atrium via a puncture wound in the left atrial appendage. The LEAP stimulus was then applied across the two catheter coils. In four of the dogs, LEAP stimulation was also delivered via patch electrodes sutured to the right and left atrial appendages. The patch electrodes consisted of 2-cm × 2-cm pieces of stainless steel wire mesh, insulated with rubber membrane on the surface not in contact with the atrial tissue. Atrial sensing was performed using a monophasic action potential (MAP) catheter (7 F, EP Technologies), which was advanced via the femoral vein into the right atrium.

The following signals were recorded: lead II surface ECG, MAP from the right atrium, bipolar electrograms from the right and left atria, and arterial blood pressure. The recordings were acquired at a sampling frequency of 1,000 Hz and stored digitally using a data acquisition system (BioPac Systems, MP 150; software, AcqKnowledge 3.7.3).

To permit stimulation of the vagus nerve and thereby facilitate the induction and maintenance of AF, the right cervical vagus nerve was isolated, doubly ligated and cut. Bipolar iridium wire electrodes, insulated except at the tip, were inserted into the sheath of the nerve and connected to a WPI stimulator. Immediately before AF induction and continuously during AF, pulses of 2-ms duration were delivered at a current strength that produced the maximum reduction in sinus rate (typically 2–4 mA) at a stimulus frequency of 10 Hz. The region surrounding the cut end of the nerve was bathed in mineral oil to prevent desiccation and preserve intact function over the 5–7-h time course of the experiment. AF was induced using a pace-down protocol, in which a train of symmetrical biphasic LEAP pulses of 8-ms duration (4 ms up/4 ms down) was delivered at an intensity of 2.0 V, initially at a cycle length of 300 ms. The cycle length was subsequently shortened progressively until AF was induced.

After instrumentation of the dog, the experimental protocol was: (1) determine the threshold for activation of the atria using a single symmetrical biphasic LEAP stimulus pulse (4 ms up/4 ms down), delivered at a constant cycle length of 400 ms; (2) determine the impedances of the LEAP electrodes by delivering single biphasic pulses of 8-ms duration at voltages of 10, 20, 40, 60, 80 and 100 V and measuring the resulting voltage deflection at the sensing electrode; (3) determine the intensity and frequency of vagal stimulation required to reduce sinus heart rate maximally; (4) induce AF using the pacedown protocol described above and monitor for 2 min; (5) attempt to cardiovert using a single electric shock of 8-ms duration, starting with a shock strength of 40 V and increasing the voltage by increments of 5–10 V until cardioversion occurred, up to a maximum of 100 V; (6a) if cardioversion was successful, reinitiate AF and attempt to suppress AF using five symmetrical biphasic LEAP pulses of 8-ms duration (4 ms up/4 ms down), delivered at a cycle length 5–10 ms shorter than the cycle length corresponding to the dominant frequency during AF, as determined using a Fast Fourier Transform (FFT) of the MAP recording, and at an initial voltage of 10 V, followed by increments of 5–10 V until AF was suppressed; (6b) if cardioversion was not successful, attempt to suppress AF using LEAP stimulation; (6c) if neither cardioversion nor LEAP stimulation was successful, turn off vagal stimulation and suppress AF using a single shock or LEAP stimulation; (7) repeat steps 4–6 for an additional 3–5 trials; (8) change the LEAP electrode configuration (for example, from catheters to patch electrodes) and repeat steps 1–7.

The experimental procedures were approved by the institutional animal care and use committee of the Center for Animal Resources and Education at Cornell University.

Cardiac preparation *in vitro*. Adult beagle dogs of either sex, age 1–4 years ($n = 5$ for the atrial and $n = 7$ for the ventricular experiments), were anaesthetized with Fatal-Plus ($390\,mg\,ml^{-1}$ pentobarbital sodium, Vortex Pharmaceuticals; 86 mg kg^{-1} intravenously) and their hearts were rapidly excised. For the atrial experiments, after excision, the right and left coronary arteries were cannulated using polyethylene tubing and the right and left atria were excised and perfused with normal Tyrode solution bubbled with 95% O_2, 5% CO_2 at p_{O2} 400–600 mm Hg, pH 7.35 ± 0.05 and temperature 37.0 ± 0.5 °C. The flow rates of the perfusate and superfusate were $20\,ml\,min^{-1}$ and $60\,ml\,min^{-1}$, respectively, at a perfusion pressure of 50–80 mm Hg. After 15–30 min of equilibration, the preparation was stained with the voltage-sensitive dye di-4-ANEPPS ($10\,\mu mol\,l^{-1}$ bolus). Blebbistatin ($10\,\mu mol\,l^{-1}$ constant infusion over 30–40 min) was added to prevent motion artefacts.

LEAP pulses were delivered from a custom-built cardioverter/defibrillator, which was capable of generating pulse sequences with a specified number of pulses, pulse duration, polarity and shape. The computer-controlled device used a digital-to-analog converter (NI USB-6259 BNC) to generate arbitrary waveforms, which were amplified using a power amplifier (Kepco BOB 100-4M). The waveform was configured manually using a Labview program or was loaded from a database. Electrophysiological signals and various monitor signals (for example, delivered current and voltage) were recorded using an analog-to-digital converter (NI USB-6259 BNC). The signals were analysed in real time and used to select the LEAP parameters automatically. An automated impedance measurement was used to calibrate the pulse energy. The data obtained during the experiment were stored in a database for offline analysis.

Standard bipolar stimulating electrodes were placed on the right and left atria (in the same positions as in the corresponding *in vivo* experiments) and field stimulation was delivered as described above. To calculate the energy delivered, the impedance of the electrodes was determined by delivering single biphasic pulses of 8-ms duration at voltages of 10, 20, 40, 60, 80 and 100 V and measuring the resulting voltage deflection at the sensing electrode. Pacing stimuli were delivered using a WPI stimulator and stimulus isolator, and LEAP stimuli were delivered using a function generator and custom-built current source. Field strengths in excess of $5\,V\,cm^{-1}$ could be delivered using this device, at cycle lengths as short as 50 ms. The field strength between the electrodes was measured using two teflon-coated silver wires immersed in the bath and set 5–10 mm apart.

To verify tissue viability, the excitation threshold for far-field stimulation in the quiescent myocardium was determined by monitoring optical wave activity after application of a sequence of five LEAP pulses, and was compared to the corresponding *in vivo* study values. AF was initiated subsequently using rapid pacing, either with or without acetylcholine (1–3 μM) in the perfusate. The concentration of acetylcholine was titrated to induce AF with a similar dominant frequency to that observed *in vivo*. The presence or absence of AF was documented by monitoring wave activity optically. The dominant frequency during AF was determined from the FFT of a single pixel recording that could be moved in real time to assess the range of frequencies throughout the tissue. At the end of the study, the excitation threshold was recalculated to confirm the stability of the tissue. For the ventricular experiments, the protocols were similar to those used for the atrial experiments, as described in detail previously[15]. However, no acetylcholine was used and fibrillation was initiated by rapid pacing.

The experimental procedures were approved by the institutional animal care and use committee of the Center for Animal Resources and Education at Cornell University.

Optical fluorescence imaging. Illumination was provided by LEDs (Luxeon, 5 W, 530 nm). High-numerical-aperture lenses (F-number 0.95, focal length 50 mm) were fitted with long-wavelength-pass emission filters (580 nm). The epicardium and endocardium were imaged simultaneously using two synchronized cameras: the control-site studies used two high-resolution, high-speed CMOS cameras (Vision Research Phantom V7, 600 × 800 pixels, 12-bit, 2,000 frames s^{-1}), whereas arrhythmia termination experiments used two EMCCD cameras (electron multiplied charge coupled device, Photometrics Cascade 128+, 128 × 128 pixels, 16-bit, 511 frames s^{-1}). The pseudo-ECG is defined as the mean of the optical fluorescence signal over the entire field of view.

Micro-computed tomography. Immediately after optical mapping experiments, tissues were injected with 1–2 ml of Microfil contrast agent (Flow Tech Inc.) at 0.05–0.15 $ml\,min^{-1}$ via the same cannula used for perfusing the tissue. The chambers were then filled with silicone to preserve tissue morphology during the scan. The contrast agent and silicone were allowed to set for at least 4 h before being scanned. The scans were performed using the GE CT120 micro-CT scanner (GE Healthcare). For each data set, 1,200 projections were obtained at 0.3° intervals over 360°, using 80 keV, 32 mA and 25 μm x–y–z resolution. Before each scan, ten bright-field images were acquired with no objects in the field of view, providing a correction for detector non-uniformity. Each image data set was transferred from the CT120 system to an image-processing workstation (HP xw8400 with 8 CPU cores and 16GB RAM). The projection views were used to reconstruct a CT image using a convolution back-projection approach implemented in three dimensions, giving a $40 \times 40 \times 40\ \mathrm{mm}^3$ volume of image data with 25 μm or 50 μm isotropic voxels in analog-to-digital units. Correction for signal non-uniformity across the field of view was determined from measurement within a water/tissue phantom (SB201), scanned with the same X-ray protocol. Ten axial slices within this water phantom were averaged to produce a two-dimensional map of offset values, used to correct for non-uniformity at any point in the image. Corrected image data sets were calibrated to the conventional scale of Hounsfield units, defined so that water and air have values of 0 and −1,000, respectively. The maximum intensity projection (MIP) shown in Fig. 3b and f is an orthographic projection which maps voxels with maximum intensity that intersect parallel rays from the viewpoint to the plane of projection. The details of the further analysis are given in Supplementary Information.